

B)	Find the Fourier series of $f(x) = 9 - x^2$ over $(-3, 3)$	Understand (CO4)	6
C)	Expand $\pi x - x^2$ as a half range Fourier cosine series for $0 \leq x \leq \pi$	Understand (CO4)	6
Q. 5	Solve any two of the following.		12
A)	If $\phi = x^2y + 2xy^2 + 3yz$ then find $\nabla \phi$ and if $\vec{F} = 2xi - yj - 2zk$ then find $\nabla \cdot \vec{F}$ and $\nabla \times \vec{F}$.	Remember (CO5)	6
B)	Evaluate $\oint_C [(xy + y^2) dx + x^2 dy]$ by Green's theorem, where C is the boundary of the region bounded by the parabola $y = x^2$ and $y = x$.	Apply (CO5)	6
C)	Use Gauss divergence theorem to evaluate $\iiint_S \vec{F} \cdot d\vec{s}$ where $\vec{F} = 4x i - 2y^2 j + z^2 k$ and S is the surface bounding the region $x^2 + y^2 = 4, z = 0$ and $z = 3$.	Apply (CO5)	6
*** End ***			